

Practical 12 - Non-linear effects and splines

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

Datasets required: `SAheart.csv`

In this practical we will explore modelling non-linear covariate effects using the same South Africa data as the previous two practicals.

Question 1

Load the data, and fit a simple logistic model for `chd` with `sbp` as the single covariate, saving the model for later comparison:

```
SAheart <- read.csv("SAheart.csv", row.names = 1)
chd_linear <- glm(chd~sbp,data=SAheart,family="binomial")
summary(chd_linear)
```

What do the model results show?

Question 2

Plot the fitted probability values from `chd_linear` against `sbp`. Can you explain what you see?

Question 3

Use the following code to generate a variable containing the loess smoother estimate of the probability of CHD at each value of SBP in the dataset:

```
chd_loess <- loess(chd~sbp,data=SAheart)
```

Next, use the `plot` function to plot `chd_loess$fitted` against `sbp`. What does the loess smoother suggest about the effect of SBP on CHD?

Question 4

Generate a new variable `loess_logit` which takes the values in `chd_loess$fitted` and converts the probabilities to log odds. Then plot `loess_logit` against `sbp`. How has the plot changed compared to the previous question?

Question 5

Fit a logistic regression for the `chd` outcome with `sbp` and the square of `sbp` as covariates. To do this you can generate a new variable in the data frame for the square term, or include `I(sbp^2)` as a covariate in your model formula.

From your fitted model what do you conclude now about the effect of SBP?

Use a plot to compare the fitted log odds values from this model with the ones estimated using `loess`.

Question 6

Use the following code to fit a logistic model for `chd` with a restricted cubic spline with 4 degrees of freedom for the effect of `sbp`:

```
library(splines)
chd_rcs <- glm(chd ~ ns(sbp,df=4),data = SAheart,family = binomial)
```

Then create a plot to compare the fitted log odds with the `loess` and quadratic model estimates.

Question 7

Re-run your code from question 6 but with either smaller values of `df` or larger values. Does changing the value of `df` change the fitted log odds in the way you would expect?

Question 8

Use the `anova` function to compare your earlier quadratic effect model with the restricted cubic spline model with `df=4`. What do you conclude?

Is it valid to compare these two models with a likelihood ratio test?

Use the `AIC` function to compare the two models with the `AIC`.

Question 9

We have throughout this practical implicitly taken the loess smoother estimate as if were ‘correct’. It is though just an estimate, and in particular the estimate it gives is affected by the bandwidth, which is specified by the `span` argument to `loess`. Examine the documentation for `loess` and re-run it with different values of `span` to see how the value affects the loess estimate of the probability of CHD by SBP.